BITS Pilani — Work Integrated Learning Programmes

M.Tech (AIML) | Machine Learning Assignment - 2 Submission

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1. GitHub Repository Link

https://github.com/Akhil4826/online-shopping-purchase-prediction

Repository contains complete source code, requirements.txt, README.md, test_data.csv, and saved models in model/.

2. Live Streamlit App Link

https://online-shopping-purchase-prediction.streamlit.app/

Interactive Streamlit web frontend deployed on Streamlit Community Cloud (evaluates held-out test data without retraining).

3. Screenshot of Assignment Execution on BITS Virtual Lab

BITS Virtual Lab Execution Screenshot Placement

Execution verification inside BITS Virtual Lab environment.
Place your screenshot as bits_lab_screenshot.png in the project directory and run python generate_submission_pdf.py to embed it directly.

4. Complete GitHub README Content

A. Problem Statement

This project predicts whether a visitor will complete a purchase during an online shopping session using browsing and session behavior data. Five classification models are trained and compared on held-out test data to see which model handles purchase prediction best.

B. Dataset Description

• **Dataset Name:** Online Shoppers Purchasing Intention Dataset
• **Source:** UCI Machine Learning Repository (ID: 468)
• **Total Records:** 12,330 sessions
• **Input Features:** 17 features (10 numerical features such as durations, bounce rates, exit rates, and page values; 7 categorical features such as month, visitor type, region, and operating system)
• **Target Variable:** Revenue (True = purchase made, False = no purchase)
• **Class Distribution:** 10,422 False (84.53%) and 1,908 True (15.47%)

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D. Models Used

1. **Logistic Regression:** A linear model predicting purchase probability using regularized log-odds.
2. **Decision Tree Classifier:** A tree-based model splitting features recursively by Gini impurity.
3. **K-Nearest Neighbors (KNN):** An instance-based classifier using distances to the 5 nearest neighbors.
4. **Naive Bayes (Gaussian):** A probabilistic classifier applying Bayes' rule assuming feature independence.
5. **Support Vector Machine (LinearSVC):** A linear classifier finding the maximum margin decision boundary.

E. Model Comparison

The table below shows the evaluation results on the 20% held-out test set (2,466 records). All metrics use Revenue = True as the positive class:

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.8808	0.8828	0.7366	0.3586	0.4824	0.4592
Decision Tree	0.8585	0.7356	0.5420	0.5576	**0.5497**	**0.4658**
KNN	0.8767	0.7633	0.6822	0.3822	0.4899	0.4493
Naive Bayes	0.2729	0.7334	0.1726	**0.9738**	0.2933	0.1289
Support Vector Machine	0.8788	**0.8960**	**0.7712**	0.3089	0.4411	0.4381

F. Observations

• **Logistic Regression:** Achieved high accuracy (88.08%) and good precision (73.66%), but had low recall (35.86%) because it predicts the positive minority class conservatively.
• **Decision Tree:** Balanced positive-class precision (54.20%) and recall (55.76%), leading to the highest F1 score (0.5497) and highest MCC (0.4658).
• **KNN:** Showed decent overall accuracy (87.67%) and precision (68.22%), but achieved modest recall (38.22%) on the minority class.
• **Naive Bayes:** Detected nearly all buyers (97.38% recall), but produced too many false positives because browsing features are correlated, dropping accuracy to 27.29%.
• **Support Vector Machine:** Achieved the highest AUC (0.8960) and highest precision (77.12%), making it the most reliable model when false positives must be avoided.

G. Overall Winner

Winner: Decision Tree Classifier
Because this dataset is imbalanced (only ~15.5% purchases), overall accuracy is dominated by the negative class. The Decision Tree is the overall winner because it achieves the highest positive-class F1 score (0.5497) and highest MCC (0.4658), identifying

over 55% of actual shoppers who buy while maintaining reasonable precision.